Ex. No.: Roll No.: 22BCS096
Date :

Component and Deployment Diagrams

Aim

To create Component and Deployment diagrams for Centralized Hostel Management system using CASE tools

Component Diagram

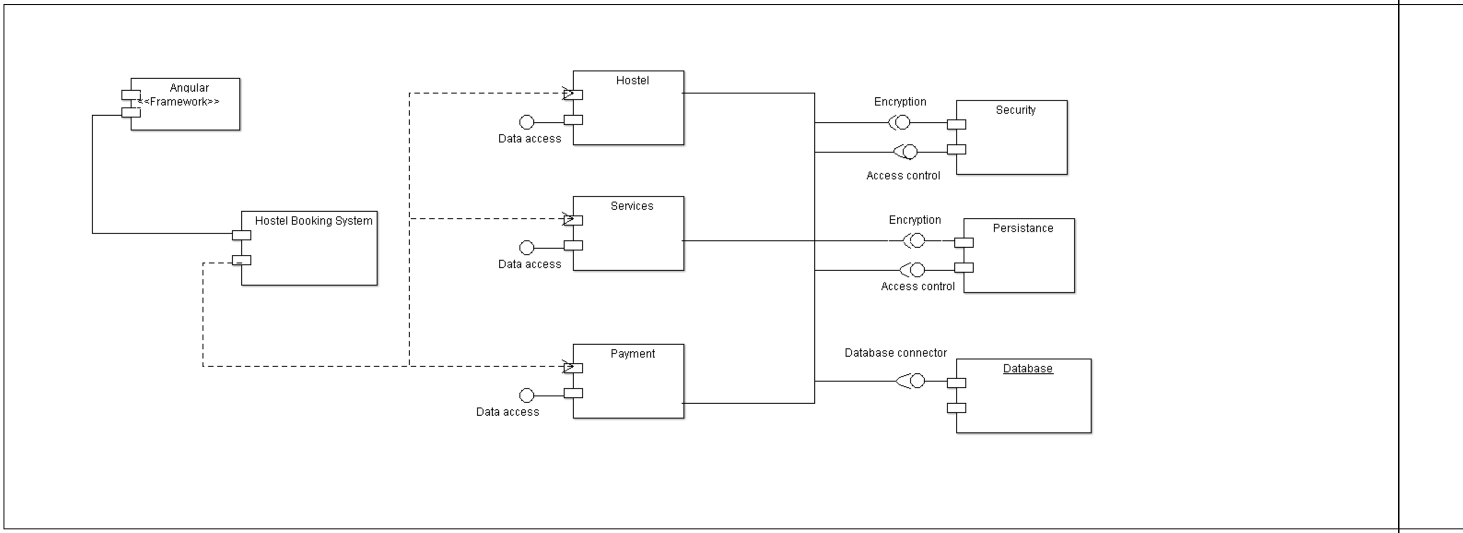

Fig: 9.1 – Component Diagram for Centralized Hostel Management system

Description

Figure 9.1 shows *the individual components of the Hostel Management System, depicted as a component diagram in Object-Oriented Analysis and Design (OOAD). This diagram illustrates the architectural structure and interactions between various components, highlighting their roles in ensuring the system's functionality.*

- ***Component 1 - Angular Framework**: This component serves as the front-end interface of the Hostel Management System. Built using the Angular framework, it provides a dynamic, user-friendly interface for users to interact with the system. It handles client-side operations, including rendering views, managing user inputs, and communicating with the back-end services via data access layers. Its purpose is to enable seamless navigation and presentation of hostel booking, payment, and management features.*

- ***Component 2 - Hostel****: This component represents the core business logic and data management module for the hostel-related operations. It encapsulates functionalities such as room allocation, booking management, and availability checks. The Hostel component interacts with the Data Access layer to retrieve and store data, ensuring that hostel-specific information is accurately processed and maintained. Its purpose is to serve as the central hub for hostel operations, coordinating with other components to fulfill user requests efficiently.*

- ***Component 3 - Hostel Booking System****: This component is responsible for managing the booking process within the system. It handles user requests for room reservations, validates availability, and processes booking confirmations. Through its Data Access connection, it updates the underlying database with booking details. Its purpose is to provide a streamlined booking experience, ensuring that users can securely and effectively reserve hostel accommodations.*

- ***Component 4 - Payment****: This component manages all financial transactions related to hostel bookings. It integrates with the Data Access layer to process payment details and update payment statuses. Its purpose is to ensure secure and reliable payment processing, supporting various payment methods and providing confirmation to users upon successful transactions.*

- ***Component 5 - Services****: This component provides auxiliary functionalities that support the core operations of the system, such as user notifications, reporting, and support services. It connects to the Data Access layer to fetch or update relevant data. Its purpose is to enhance the user experience by offering additional services that complement booking and payment processes.*

- ***Component 6 - Security****: This component handles the security aspects of the system, including encryption and access control. It ensures that data transmitted between components and stored in the database is encrypted, protecting sensitive information. Its purpose is to safeguard the system against unauthorized access and ensure compliance with security standards.*

- ***Component 7 - Persistence****: This component manages data persistence and storage within the system. It includes the Database Connector, which facilitates interactions between the application and the Database. Its purpose is to ensure that all transactional data (e.g., bookings, payments) is reliably stored and retrieved, maintaining data integrity and availability.*

- ***Component 8 - Database****: This component serves as the centralized data repository for the Hostel Management System. It stores all relevant data, including user information, booking* *details, and payment records. Accessed via the Database Connector, its purpose is to provide a robust and scalable storage solution that supports the system's operational needs.*

Deployment Diagram

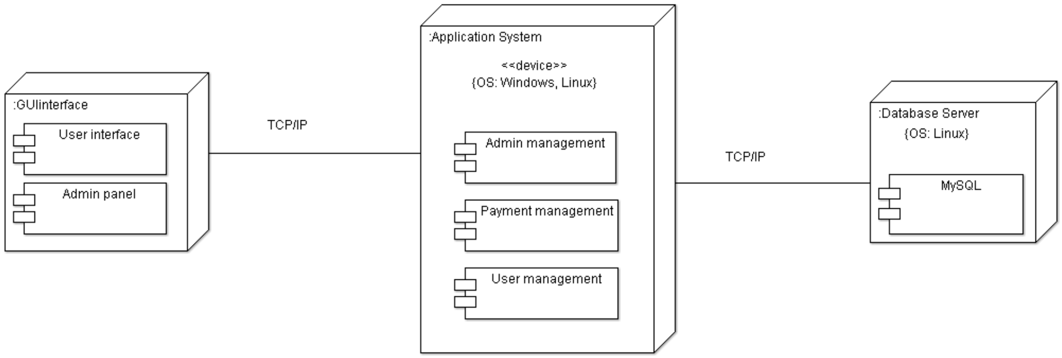

Fig: 9.2 – Deployment Diagram for Centralized Hostel Management system

Description

Figure 9.2 *shows the deployment diagram of the Application System, illustrating the physical deployment of the software components across various devices and their communication over a network. This diagram provides a high-level view of the system's architecture, highlighting how the software is distributed and interacts with the underlying hardware and network infrastructure.*

- ***Device Nodes** a. **Device 1 – GUI Interface**: This device represents the client-side node where the user interacts with the Application System. It hosts the GUI Interface, which includes the User Interface and Admin Panel components. The purpose of this device is to provide a graphical front-end for users to manage administrative tasks, perform payment operations, and handle user-related activities. Operating on a TCP/IP network, it connects to the Application System server, enabling seamless data exchange for real-time updates and user actions. This node is typically a workstation or personal computer running an operating system such as Windows or Linux.*

*b. **Device 2 – Application System**: This device serves as the central server node that hosts the core application logic of the system. It runs on an operating system such as Windows or Linux and contains the Admin Management, Payment Management, and User Management components. The purpose of this device is to process and manage the business logic, handle payment transactions, and maintain user data. It communicates with the GUI Interface via TCP/IP and interacts with the Database Server to store and retrieve data, ensuring the system's operational integrity and scalability.*

*c. **Device 3 – Database Server**: This device acts as the data storage node, hosting the MySQL database. Operating on a Linux operating system, its purpose is to provide a reliable and efficient storage solution for all system data, including user information, payment records, and administrative details. It connects to the Application System via TCP/IP, facilitating data persistence and retrieval to support the system's functionality. This node ensures data integrity and availability, serving as the backbone for the application's data management needs.*

The deployment diagram depicts the communication between these device nodes using TCP/IP protocols, emphasizing a distributed architecture. The GUI Interface device initiates requests to the Application System, which in turn queries the Database Server for data operations. This structure ensures a modular and scalable deployment, with each device fulfilling a specific role to support the overall functionality of the Application System.

Result

Thus the component and deployment diagrams for Centralized Hostel Management system is created using CASE tools.